

26. FRUIT AND ITS COLOR USING BACKTRACKING

Aim

To write a Prolog program to identify the color of a fruit using facts, rules, and backtracking.

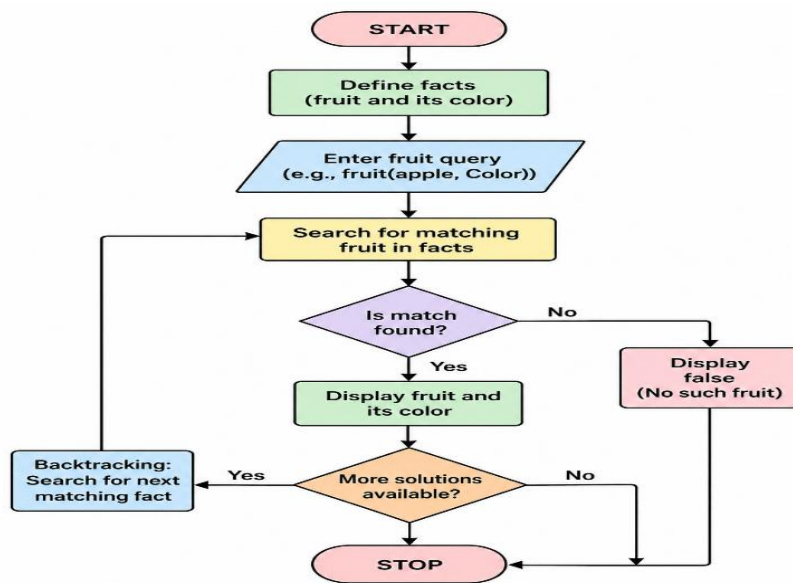
Objective

- To understand the basic concepts of Prolog.
- To represent fruits and their colors using facts.
- To use rules for querying fruit colors.
- To understand how Prolog uses backtracking to find multiple solutions.

Algorithm

1. Start the program.
2. Define facts representing fruits and their colors.
3. Define a rule fruit_color/2 to find the color of a given fruit.
4. Enter a query with a fruit name.
5. Prolog searches the facts from top to bottom.
6. If a matching fact is found, Prolog returns the corresponding color.
7. If the user requests another solution using ;, Prolog performs **backtracking** and searches for another matching fact.
8. Display the result.
9. Stop.

Flowchart



Prolog Code

fruit(apple, red).

fruit(banana, yellow).

fruit(grapes, green).

fruit(orange, orange).

fruit(cherry, red).

Output:

```
SWI-Prolog (AMD64, Multi-threaded, version 9.2.9)
File Edit Settings Run Debug Help
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

?-
% c:/Users/Admin/OneDrive/Documents/fruit and color using backtracking.pl compiled 0.00 sec, 5 clauses
?- fruit(X,red).
X = apple ,

?- fruit(grapes,X).
X = green.

?-
```

Result

Thus, the Prolog program to identify the color of a fruit using backtracking was successfully implemented and executed. Prolog returns the possible colors one by one, demonstrating the concept of backtracking.